Calculating safety & stress

Yield strength for Grade A A500 steel Round structural tubing: 33,000 psi, 228 MPa.

$S_y = 228\,MPa$

Safety Factor: $SF = 3.5$

$F_{max} = 20.031\,KN$

$\sigma = \frac{S_y}{SF} \Rightarrow \frac{F}{A_{min}} = \frac{S_y}{SF} \Rightarrow A_{min} = F \cdot \frac{SF}{S_y}$

$$A_{min} = \frac{F(SF)}{S_y} = \frac{(20.031 \cdot 10^3\,\cancel{N})(3.5)}{228\,\frac{\cancel{N}}{mm^2}} = \boxed{307.49\,mm^2}$$

Area of a circle

$A = \pi r^2$

$\pi r^2 = 307.49\,mm^2$

$r^2 = \frac{307.49\,mm^2}{\pi}$

$r = (+)\sqrt{\frac{307.49\,mm^2}{\pi}}$

$r = 9.8933\,mm$

Rounding up

~~$r = 9.9\,mm$~~

$r = 10\,mm$ rounding up to 10 for easier calculations

~~$d = 2r = 2(9.9) = \boxed{19.8\,mm}$~~

$d = 2r = 2(10) = \boxed{20\,mm}$